Midterm Instructions

Data Processing in Python (JEM207)

General Guidelines

Dear students, please read this carefully.

The midterm exam is designed to test your ability to **think like a programmer**, handle real-world data processing problems, and break tasks into logical steps.

You will most likely run into challenges — do not give up! Coding requires patience and experience. Approach the exam with a cool head, use your reasoning, and take advantage of Google and Stack Overflow for troubleshooting.

Important:

- It is **prohibited to share code** with other students.
- It is **strictly prohibited to use generative AI tools** such as ChatGPT.

Instructions

- Make sure you know your **CUNI student number** (as in your email, e.g., `12345678@fsv.cuni.cz`).
- **Deadline:** April 13, 2026 at 12:20.
- Work preferably in **Jupyter Lab** or **Jupyter Notebook**.
- Submit your work as a folder containing a Jupyter notebook file (`.ipynb`) and data, in a GitHub repository.
- Register your repository using this form: Google Form. You can do this after finishing the exam, but there should be no unreasonable delay between the deadline and the form submission.
- Only commits pushed before the deadline will be considered.
- Your code must run **from scratch without errors**, and **output** the required solutions.
- The midterm is worth **25 points in total**.
- Solve each problem in a separate cell. Clearly indicate which problem you are solving.
- The exam consists of three parts.
- The time might be tight. If you get stuck proceed to the other tasks and come back if you have time left. You will be able to complete most of the tasks without having completed the previous ones.

- In this case (and this case only) concentrate on the correctness of the outputs, rather than design of code and outputs.

If you show a reasonable attempt, you can still receive partial credit. **Happy coding!**

Data Analysis (25 points total)

It is suggested to create a `pandas` **DataFrame** with dates as the index.

Part 1 (14 points)

Get the data (0 points):

- Download your custom dataset at Google Drive. Your dataset will be under `your_student_cuni_number.zip`.
- If you cannot find your file, contact the instructor.
- It is a simple ZIP file. You can extract it manually using your operating system tools.
- Ensure you can access the CSV files in your Jupyter notebook.
- Each file name corresponds to a company ticker — do not rename it.
- Create a Ticker column identifying each of the companies.
- Ensure that Dates are correctly represented as `datetime` objects.

Use the **Close** time series for the following analysis unless specified otherwise.
Tasks:

1. (2 pt) Is there a company that has no difference between the High and Low columns for any trading day? What does it mean for the stock's trading behavior on such days?
2. (1 pt) What is the highest and lowest price price each company recorded? *Hint:* Select the column(s) that can contain the maximum and minimum price wisely.
3. (1 pt) Calculate logarithmic returns from **Close**. For each company, report min, max, mean, standard deviation, and skewness of returns.
4. (2 pt) When (date) did each company record the second highest gain and second highest loss (in terms of logarithmic returns)? *Hint:* `idxmax`.
5. (2 pt) What is the median calendar weekly volume (i.e. sum of volume for a particular week) for each company? *Hint:* check `resample`.
6. (2 pt) Which company recorded the lowest total logarithmic return over the whole period?
7. (2 pt) Create a new column `volume_class` with categories (e.g., "Low", "Medium", "High") based on volume tertiles within the given company.

8. (2 pt) (A) Plot the log-returns of the companies (ideally in the same plot). (B) Plot the distribution of log-returns (ideally in the same plot).

See Part 2 on the next page.

Part 2 (9 points)

Download the dataset of S&P 500 companies under `companies_no_subindustry.csv` in the `companies` folder on the Google Drive.

Tasks:

1. (1 pt) How many companies have missing inclusion dates (`included` column)?
2. (2 pt) Remove companies with missing inclusion dates. Find the oldest and youngest constituent (in terms of inclusion date), and compute the median age of a constituent in years (median difference in years between today's date and date `Founded`). The format of the values in `Founded` column differs. You can unify it by taking the first **4 characters** of each entry.
   Hint: `pd.to_datetime(..., dayfirst=True, errors='coerce')`.
3. (2 pt) Compute the number of companies in each of the sectors and visualize proportions (e.g., pie chart).
4. (2 pt) Parse the `hq` column to extract the state and describe its distribution (e.g., bar plot).
5. (2 pt) Join this dataset with `companies_subindustry.csv` that you will also fond in the `companies` folder on the Google Drive. Merge based on an appropriate key and report subindustry distribution within the "Health Care" GICS sector (e.g., bar plot).

Part 3 (2 points)

1. (1 pt) Generate four random variables from four different distributions (e.g., Normal, Uniform, etc.), each with a random number of observations between 100 and 500.
2. (1 pt) Draw a matrix of plots (e.g., PDF, CDF, histogram, or boxplot) visualizing the random variables.

Submission Checkpoints

- **Save your work!**
- Ensure your notebook (`.ipynb`) runs from scratch — restart the kernel, re-import everything, and verify it works.
- Commit and push your work to GitHub (only pushed commits count). You may also upload directly in Github.

Useful formulas

$$r_t = log(P_t) - log(P_{t-1}), \tag{1}$$

where r_t is the logarithmic return for day t, log() indicates the natural logarithm, and P_t indicates the price for day t.

$$r_T = log(P_T) - log(P_0) = \sum_{t=1}^{T} r_t, \tag{2}$$

where r_T is the **Total logarithmic return** from day 0 to day T.